Zero branches preserve the entire class of causal seed laws

Research note for the finite-Shannon-theorem project

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Abstract

For a locally finite controlled tree with nonnegative normalized kernels, we prove that positive-support reduction preserves exactly which probability spaces can serve as causal seeds, simultaneously at every finite horizon. The seed is unchanged; only the decoder is transformed. The analogous statement holds for ordinary policy-indexed sources. Consequently the positive-kernel infinite-horizon theorems transfer to the original tree, with its zero-probability branches retained and its decoder total there. This is a mathematical extension, not a new Lean certificate or a claim of historical priority. The underlying support-reduction idea is standard.

1 Model and an exact equivalence

At each finite history h, let $A(h)$ be a finite nonempty action set; for $a \in A(h)$ let $Y(h,a)$ be a finite nonempty output set, with

$$p(y \mid h,a) \geq 0, \qquad \sum_{y \in Y(h,a)} p(y \mid h,a) = 1.$$

Neither stationarity nor a uniform positive lower bound is assumed. A terminal history, if present, can be continued by a deterministic dummy action and output. The set of finite histories is countable.

Call a history *positive* if all transition probabilities along it are positive, and let H^+ be their set, including the empty history. The auxiliary tree T^+ keeps the same actions at histories in H^+ and replaces $Y(h,a)$ by $Y^+(h,a) = \{y : p(y \mid h,a) > 0\}$. Its transition probabilities are unchanged: there is no renormalization. Write j for inclusion of its histories into those of the original tree T.

On a probability space $(\Omega, \mathcal{F}, \mu)$, a causal decoder is a measurable response $D(\omega, h, a) \in Y(h,a)$ for every history and action. It is exact if every deterministic adaptive policy has the prescribed finite transcript laws. Measurability means measurability of each finite output coordinate. An ordinary source instead has separate measurable transcript readouts for each finite horizon and policy, with the same exact marginals and without consistency requirements between readouts.

Theorem 1 (Preservation of the seed space)**.** *For every probability space $(\Omega, \mathcal{F}, \mu)$,*

$$\boxed{(\Omega,\mu) \text{ admits an exact causal decoder for } T \iff (\Omega,\mu) \text{ admits one for } T^+.}$$

The equivalence holds both at any fixed horizon and simultaneously at all finite horizons. It also holds with "causal decoder" replaced by "ordinary source." All conversions leave Ω, $\mathcal{F}$, and μ unchanged. On total strategy spaces the extension map is an injective measurable section of the restriction map.

Proof. Choose a positive default output $y_*(h,a) \in Y^+(h,a)$ at every original history and action; normalization ensures existence. Put

$$r_{h,a}(y) = \begin{cases} y, & p(y \mid h,a) > 0, \\ y_*(h,a), & p(y \mid h,a) = 0. \end{cases}$$

For a full decoder D define its reduced decoder by

$$RD(\omega, h, a) = r_{j(h),a}(D(\omega, j(h), a)).$$

Conversely, for a reduced decoder D^+, define a *total* full decoder

$$ED^+(\omega, h, a) = \begin{cases} D^+(\omega, j^{-1}(h), a), & h \in H^+, \\ y_*(h, a), & h \notin H^+. \end{cases}$$

Both maps are measurable coordinatewise, and $RED^+ = D^+$ pointwise. In particular E is injective. The definitions do not depend on a horizon.

Suppose first that D^+ is exact. For any full policy π, restrict its choices to positive histories, obtaining π^+. Running ED^+ under π stays in H^+ for *every* seed value and is precisely the included run of D^+ under π^+. Under the prescribed full law, any path leaving H^+ has probability zero, while every positive path has its unchanged product probability. Thus ED^+ is exact.

Now suppose D is exact. Extend a reduced policy π^+ to a full policy π by arbitrary legal choices outside H^+. For each finite horizon, the D-run under π has probability zero of taking a zero transition, by its exact transcript law. Until such a transition, the D-run and the RD-run coincide. Therefore the reduced run has exactly the required law. This proves exactness at every horizon using the single decoder RD above; it is not an argument that chooses separate seeds or decoders for each horizon.

There is also a common full-measure set on which D never chooses a zero output at any history reachable under any policy. For each finite path ending in a zero transition, consider the event that D agrees with all its specified outputs. A policy that follows the specified actions on that path shows that this event has measure zero. There are countably many such paths, so their union is null. No uncountable union over infinite policies is needed. This statement concerns reachable histories, not arbitrary unused coordinates of a strategy.

For ordinary sources, reduced-to-full conversion restricts a full policy, uses its reduced readout, and includes the resulting transcript. In the other direction, extend the reduced policy and use the full readout. A positive transcript is restricted; replace a nonpositive one by any fixed positive transcript compatible with the reduced policy. The replacement event has measure zero, and the readout remains measurable. This construction works separately at each readout index, exactly as the definition of an ordinary source permits. Throughout, the seed law is literally unchanged. □

2 Spectra and the infinite consequences

For a fixed policy and horizon, support reduction preserves the multiset of positive transcript masses. Every full policy restricts and every reduced policy extends. Consequently, for every n, t, K,

$$F^*_{T_n}(t) = F^*_{T_n^+}(t), \qquad \inf_\pi A_K(P^T_{n,\pi}) = \inf_\pi A_K(P^{T^+}_{n,\pi}). \tag{1}$$

Here A_K is the sum of the K largest masses, padded with zeros if necessary. Information distributions sum over positive transcript masses. Zero-mass leaves contribute no probability at $+\infty$.

Corollary 2. *The positive-tree infinite-horizon results extend to T with all zero branches retained. In particular:*

1. *For countable-seed Shannon cost, in a single consistent unit,*

$$C_\infty(T) = \sup_n C_n(T).$$

If finite, the optimum is attained by a countable seed on the original tree.

2. *In bits, with* $L_\infty = \sup_n L(T_n)$ *and* $c_* = (1 + \log_2 e)/2$,

$$L_\infty \le C_\infty(T) \le L_\infty + c_*.$$

An exact finite-entropy seed exists if and only if $L_\infty < \infty$.

3. *For sources with measurable singletons, define* $q = \sup_K \inf_{n,\pi} A_K(P^T_{n,\pi})$ *and* $F_\infty(t) = \inf_n F^*_{T_n}(t)$. *The maximal ordinary and causal atomic capacities equal* q, *the causal capacity is attained, and*

$$q = \lim_{t\to\infty} F_\infty(t) = \sup_t F_\infty(t).$$

An exact countably supported causal seed exists if and only if $q = 1$.

4. *If a countable ordinary exact source exists, there is one countable causal seed* S, *exact at all horizons, for which simultaneously for all* $\alpha > 0$,

$$R_\alpha(Z) \le O_\alpha(T) \le C_\alpha(T) \le H_\alpha(S) \le R_\alpha(Z) + d_\alpha,$$

with the same constants as for the positive tree. Its min-entropy attains both the ordinary and causal optima. Here Z *is the envelope-limit law, including its possible mass* $1 - q$ *at* $+\infty$.

Proof. Theorem 1 identifies, at every finite horizon and at the infinite horizon, the sets of admissible countable seed probability laws. Thus Shannon costs and all ordinary and causal Rényi costs agree with those of T^+. Entropy attainment and the simultaneous choice of a seed transfer by E, without altering that seed's weights. Equation (1) gives equality of L_∞, F_∞, q, and the envelope-limit law, so all the displayed spectral bounds follow from the positive-tree theorems.

For atomic capacity, apply the ordinary upper bound to the unchanged source space using R. To transfer attainment as a measure on total strategies, push the positive-tree attaining measure through E. Because E is injective, every target singleton has as preimage either one singleton or the empty set; total singleton mass is preserved. The pushed-forward law is exact by Theorem 1. This also justifies attainment in the strategy-space formulation, rather than only on an abstract seed space. □

3 Scope, dependencies, and what remains open

The extension retains a total decoder on the original tree. The positive tree is an auxiliary proof device, not a replacement of the final model. No limiting perturbation $p \mapsto p + \varepsilon$ is used, so there is no interchange of a positivity limit with an infinite horizon.

The input results are Theorems 5–8 of the repository README and its atomic spectral identity; the new mathematical obligation is entirely Theorem 1 and (1). Relevant Lean modules are `EntropyLimit`, `InfiniteEnvelope`, `AtomicCapacity`, `AtomicSpectrum`, and `RenyiLimitMeasure`. The repository already has finite stationary support reduction in `ZeroKernel` and `ZeroCost`; this note supplies the general horizon-independent argument for history-dependent trees. It has not been formalized or compiled in Lean.

A stronger claim that adjoining arbitrary strictly lower-entropy actions changes every C_n by at most a constant is *not proved here*. The bounded-excess theorem for a unique maximizing action does not alone establish that claim when several actions maximize entropy. The exact G_*, endpoint minimizers, optimal overhead constants, and the equal-variance next-order problem remain unresolved by these arguments.